

Article

System Design of a Low-Power BLE Smart Label SoC with Dynamic E-Paper for QR Rendering and Temperature Sensing

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Abstract

Smart labels are emerging as a key enabling technology for product traceability, environmental monitoring, and user interaction within Internet of Things (IoT) ecosystems. This work presents the design and experimental validation of a low-power smart label platform integrating Bluetooth Low Energy (BLE) communication, temperature sensing, and dynamic e-paper visualization based on the HY0020 System-on-Chip (SoC). This platform was implemented on a custom Printed Circuit Board (PCB) designed around a 1.02-inch monochrome e-paper display and incorporates a TXS0108E interface to support reliable display communication. The developed prototype enables wireless user interaction, dynamic QR code rendering, and ambient temperature monitoring while maintaining low average power consumption. Experimental evaluation included BLE communication testing, display operation validation, temperature monitoring assessment using the integrated HY0020 sensor, and energy consumption characterization. Experimental results confirmed reliable BLE connectivity, stable temperature monitoring performance under normal environmental conditions, and an estimated battery lifetime of approximately 54 days under the evaluated operating profile. The presented platform demonstrates the feasibility of integrating sensing, wireless communication, and electrophoretic display technology within a compact battery-powered smart label device. The proposed architecture provides a practical proof-of-concept foundation for future applications involving product traceability, digital information management, and Digital Product Passport (DPP)-oriented services.

Keywords: smart labels; BLE; e-paper display; temperature sensing; Digital Product Passport; low-power embedded systems; QR codes



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1. Introduction

The increasing adoption of Internet of Things (IoT) technologies has accelerated the development of connected solutions for product traceability, condition monitoring, and user interaction across sectors such as retail, logistics, healthcare, and food distribution [1,2]. Recent regulatory initiatives, including the Digital Product Passport (DPP), have further reinforced the need for digital systems capable of providing accessible and up-to-date product information throughout the product lifecycle [3]. These developments are closely aligned with broader sustainability objectives, including those promoted by the United Nations Sustainable Development Goals (SDGs) [4].

Product identification is commonly achieved through technologies such as Radio-Frequency Identification (RFID) and Quick Response (QR) codes [5]. Although both approaches are widely adopted, they primarily provide identification and information access functions. Conventional implementations offer limited support for dynamic information updates, environmental sensing, or direct interaction with end users.

Recent advances in low-power embedded systems have enabled the development of smart labels that combine sensing, wireless communication, and local information display within compact battery-powered platforms. Such systems can provide real-time status information while maintaining low energy consumption and small physical dimensions. For indoor applications, Bluetooth Low Energy (BLE) is particularly attractive because of its low power requirements, native smartphone compatibility, and widespread availability. By comparison, Long Range Wide Area Network (LoRaWAN) and Narrowband Internet of Things (NB-IoT) are generally intended for long-range infrastructure-oriented deployments, while Wi-Fi typically requires higher energy consumption [6–8].

At the same time, electronic paper displays have emerged as suitable technology for battery-powered information systems due to their ability to retain displayed content without continuous power consumption [9–11]. This characteristic makes them particularly attractive for smart label applications where information changes infrequently but must remain continuously visible.

The application scenario considered in this work is illustrated in Figure 1. The proposed platform is intended as a smart label capable of combining wireless communication, environmental monitoring, and dynamic information presentation within a compact form factor. Potential application areas include product traceability, retail systems, and DPP services.

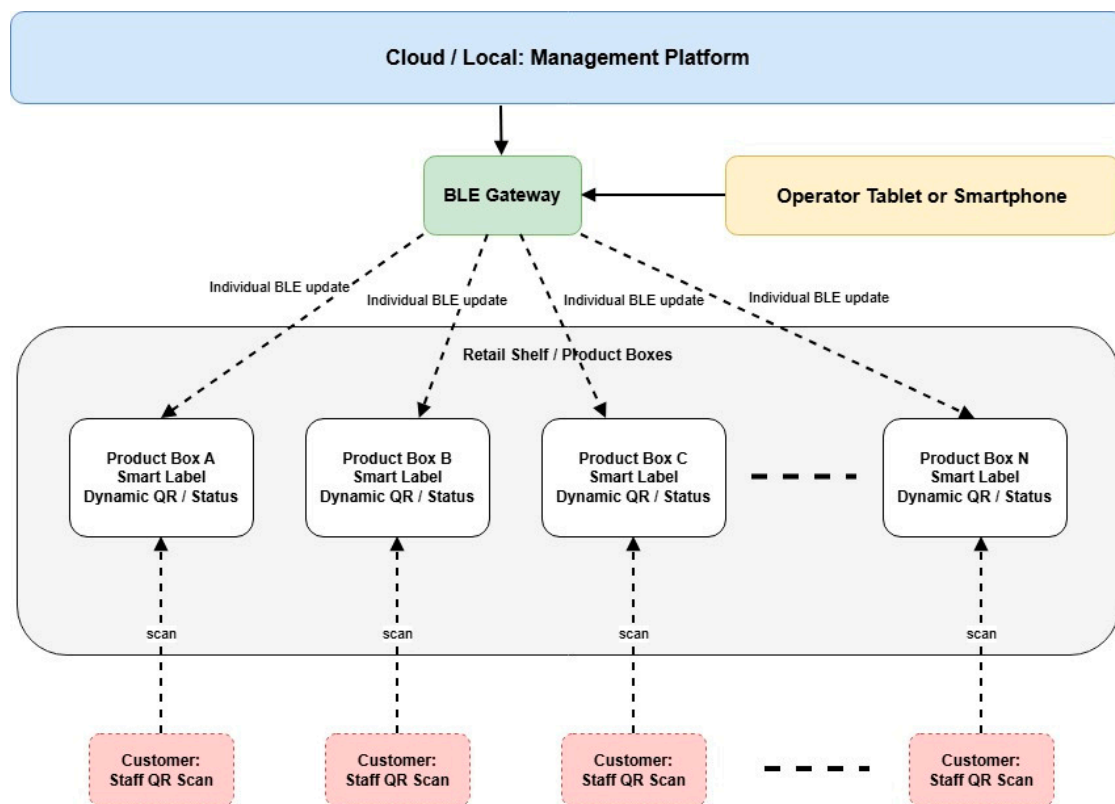


Figure 1. Example application scenario in a smart retail environment, where multiple product units are equipped with BLE-enabled smart labels with dynamic e-paper displays. A gateway can communicate individually with each label to update QR codes, product information, or status data in real time.

This article presents the design and experimental evaluation of a compact smart label platform integrating BLE communication, temperature sensing, and dynamic e-paper visualization. The system was implemented using a custom printed circuit board and evaluated with respect to communication functionality, sensing behavior, display operation, and energy consumption.

The main contributions of this work are:

1. Development of a compact smart label architecture integrating BLE communication, temperature sensing, and e-paper visualization;
2. Implementation of dynamic QR code rendering on a low-power electrophoretic display;
3. Integration of sensing, communication, and display subsystems within a custom embedded platform;
4. Experimental evaluation of communication performance, sensing behavior, and energy consumption;
5. Assessment of the feasibility of battery-powered operation for smart label applications.

The remainder of this article is organized as follows. Section 2 presents the background concepts and reviews the related literature. Section 3 describes the hardware architecture, firmware implementation, and experimental methodology. Section 4 presents experimental results and system evaluation. Section 5 discusses the main findings, limitations, and implications of the proposed approach. Finally, Section 6 summarizes the conclusions and outlines directions for future work.

2. Background and Related Works

This section provides the theoretical and technological background necessary to understand the proposed smart label platform and reviews the most relevant developments reported in the literature. Smart label systems combine several technological domains, including embedded electronics, wireless communication, sensing, and low-power display technologies. Consequently, the design of such systems requires a multidisciplinary perspective that considers both hardware constraints and application requirements.

Modern smart labels extend the functionality of conventional identification technologies by incorporating sensing, communication, and information display capabilities within compact embedded platforms. Their development is typically constrained by limited energy resources, reduced physical dimensions, and long operational lifetimes. As a result, the selection and integration of enabling technologies play a central role in the overall system design.

Section 2.1 introduces the fundamental concepts relevant to this work, including RFID and BLE communication technologies, electrophoretic (e-paper) displays, and ultra-low-power System-on-Chip (SoC) devices. Attention is given to the characteristics that make these technologies suitable for battery-powered smart label applications. Section 2.2 reviews representative studies on e-paper-based systems and smart label technologies, with emphasis on architectures that combine wireless communication, sensing, and low-power information display. The reviewed literature is analyzed in terms of system integration, communication capabilities, sensing functionality, and suitability for practical deployment. Based on this review, Section 2.3 identifies the main challenges that remain insufficiently addressed in current implementations. Rather than focusing on individual technological components, the proposed work considers the integration of sensing, wireless communication, and dynamic visual information within a compact low-power platform. This analysis provides motivation for the system architecture presented in the following sections.

2.1. Background: Smart Labels, RFID, BLE, E-Paper, and HY0020-Based Low-Power Operation

Smart labels extend conventional product identification by integrating sensing, wireless communication, processing, and local information display within compact embedded platforms [1]. Such systems support product traceability, condition monitoring, and user interaction throughout the product lifecycle. These systems are increasingly used to support product traceability, condition monitoring, and customer-facing information throughout the supply chain.

RFID represents one of the foundational technologies in this domain. RFID remains one of the most widely adopted technologies for product identification due to its low cost, scalability, and compatibility with existing logistics infrastructures [5]. RFID enables item-level identification and inventory management and can support sensing functions when combined with dedicated sensor-enabled tags. The design of RFID labels focuses on machine-readable content which allows products to be scanned by machines instead of showing information to people through continuous visual displays. RFID technology enables efficient identification and backend tracking but does not provide persistent visual information directly on the product.

Selection of communication technology for the battery-powered label needs to match the specific requirements of its application. LoRaWAN and NB-IoT are both highly relevant IoT options, especially when long-range connectivity and large-area deployment are required [6]. Both technologies are primarily intended for large-scale and long-range IoT deployments, making them less suitable for short-range product label applications. NB-IoT needs cellular network infrastructure for operation which leads to higher energy expenses, while LoRaWAN provides energy-efficient performance for low-rate long-distance data transmission instead of supporting direct short-range product interaction. Indoor environments such as retail shelves, warehouses, laboratories, and storage facilities typically require short-range, low-power communication technologies optimized for direct user interaction and local data exchange.

Wi-Fi, by contrast, is readily available in indoor environments and could support direct Internet Protocol (IP)-based communication. Wi-Fi usage in small battery-operated embedded devices results in elevated energy expenditure because of its extensive communication requirements and complete communication system. BLE is particularly attractive for smart label applications because it combines low power consumption, native smartphone compatibility, and sufficient throughput for configuration, sensing, and status-reporting tasks [7]. Experimental studies have shown that BLE power consumption depends strongly on connection parameters and device roles. Nevertheless, BLE remains one of the most energy-efficient wireless technologies for battery-powered IoT devices requiring long operational lifetimes. Smart labels which need mobile device provisioning and sensor data exchange with status updates should use BLE because of its efficient operation.

E-paper technology differs fundamentally from emissive display technologies because it relies on electrophoretic particle movement rather than light emission. Electrophoretic e-paper displays use charged pigment particles suspended in microcapsules or microcells; when an electric field is applied, these particles move and modify the visible reflectance of each pixel [12]. In monochrome displays, dark and light pigment particles move in opposite directions under the influence of the applied electric field, generating visible black and white pixel states. The system generates its output through reflective mechanisms which enable users to read content under natural light without needing backlight support.

One of the most important characteristics of e-paper technology is its bistable operation. Displayed content remains visible after the driving field is removed because the particles have reached their required optical state [12]. Display energy consumption occurs only during refresh events according to Liquid Crystal Display (LCD) and Organic

Light-Emitting Diode (OLED) panels, while their power usage remains inactive throughout display operation. A first-order update energy model can be written as

$$E_{\text{update}} = \int_0^{T_{\text{refresh}}} V(t)I(t) dt, \quad (1)$$

where

- E_{update} : Energy consumed during a display update cycle.
- T_{refresh} : Total duration of the e-paper refresh process.
- $V(t)$: Instantaneous supply voltage applied to the display at time t .
- $I(t)$: Instantaneous current drawn by the display during the refresh at time t .
- t : Time variable over the refresh interval.

The formulation highlights that the energy consumption of electrophoretic displays, commonly known as e-paper displays, is dominated by refresh operations. Power is primarily consumed during state transitions, whereas static display periods require negligible energy. The average value is given by Equation (2):

$$E_{\text{update}} \approx V_{\text{EPD}} I_{\text{avg}} T_{\text{refresh}}, \quad (2)$$

where T_{refresh} is the update time, V_{EPD} is supply voltage of the e-paper display, and I_{avg} is average current drawn by the display during the refresh process. This approximation assumes quasi-constant voltage and average current during the refresh interval, which is valid for system-level energy estimation and practical measurements in embedded implementations. Average display power over a window T can then be approximated as

$$P_{\text{avg,EPD}} \approx \frac{N_{\text{updates}} E_{\text{update}}}{T}. \quad (3)$$

The parameter N_{updates} directly reflects the display update frequency and therefore plays a significant role in determining overall energy consumption. Minimizing unnecessary updates is essential for maximizing battery lifetime in smart label applications. This formulation makes explicit why e-paper is advantageous for labels: if the label content changes infrequently, the long static intervals contribute very little to average energy consumption.

The operation of e-paper displays is strongly influenced by the waveform sequences used to control particle movement within the display medium. Unlike conventional LCD or OLED technologies, e-paper displays require carefully designed refresh procedures to achieve adequate contrast, minimize image retention, and reduce ghosting effects [12]. The refresh process is therefore a relevant system-level consideration, particularly in low-power applications where display updates must be balanced against energy consumption and user responsiveness.

The Waveshare 1.02-inch e-paper module employed in this work requires approximately 2 s for a full display refresh [13]. The display driver employs full-screen refresh operations. Although the selected e-paper module supports partial refresh functionality, this feature was not used in the present implementation. Full refreshes were preferred to ensure consistent image quality and minimize ghosting effects during repeated updates. Display waveform management is handled internally by the display controller and associated vendor library. Although this refreshing time is significantly longer than that of conventional displays, the bistable nature of the technology allows displayed information to remain visible without continuous power consumption. Consequently, display update frequency becomes an important design parameter, influencing both perceived responsiveness and overall energy consumption. These characteristics must be considered during

firmware development, particularly when integrating dynamic content such as QR codes and user-generated information.

Smart label system requires its device to use two operations—controller and radio duty cycling—together with sparse display updates. A system-level average power model can be expressed through a basic equation which describes the system's average power consumption:

$$P_{\text{avg,sys}} \approx P_{\text{sleep}} + \frac{N_s E_s + N_{BLE} E_{BLE} + N_u E_{\text{update}}}{T}, \quad (4)$$

where E_s is energy per sensing event, E_{BLE} is energy per BLE exchange, and E_{update} is the display update energy. This decomposition highlights that device lifetime depends not only on display technology but also on the coordination of sensing, communication, and visualization activities. The benefits of e-paper technology will be diminished when the radio device activates excessively, or the display system undergoes unnecessary updates.

The HY0020 platform [8–10] constitutes the technological foundation of the proposed system. In [14], the authors demonstrated how HY0020 functions as a compact IoT temperature monitoring device which requires low power while connecting through BLE and executing embedded processing and temperature sensing on a custom PCB design. HY0020 demonstrated its ability to maintain extremely low average current consumption while delivering continuous monitoring capabilities and operating through extended periods from a coin-cell-powered custom board. The system provides a solid foundation which enables development of an e-paper-based smart label system.

Reference [14] demonstrated the relevance of temperature sensing for traceability-oriented smart label applications. Temperature stands as the primary environmental factor that affects logistics operations, food quality maintenance, and the handling of products that require specific conditions. Energy representation of a sensing event can be expressed through the following equation:

$$E_{\text{sense}} = V_s I_s T_s, \quad (5)$$

where V_s , I_s , and T_s are sensing voltage, current, and acquisition time. In low-power systems, the objective is not merely to reduce E_{sense} , but to coordinate sensing with radio and display activity so that wake-up overhead is amortized across useful operations.

The present work builds upon the HY0020-based temperature monitoring platform reported in [14], extending the architecture with e-paper visualization and smart label functionality. The proposed architecture follows a hardware co-design approach in which sensing, communication, display, and power management functions are integrated within a single PCB platform.

Such integration requires more than the selection of a controller and display module. Schematics establish a dedicated interface design, which includes the TXS0108E logic-level converter [11] and all e-paper control signals (BUSY RST DC CS CLK DIN) and necessary power stage components. Integration serves as an essential design element which enables reliable communication between different voltage levels and ensures proper functioning of the display.

Practical e-paper smart label implementations face three main hardware challenges [14]: (i) mixed-voltage interfacing between the SoC and display, (ii) correct timing and sequencing of display control signals, and (iii) strict size constraints associated with label-oriented form factors.

2.2. Related Works on E-Paper-Based Smart Labeling and Display Systems

Table 1 summarizes representative studies related to e-paper displays, wireless information systems, and smart label technologies. The comparison considers the application context, communication and sensing capabilities, display functionality, hardware integra-

tion, and power-related characteristics in order to position the proposed platform with respect to existing approaches. Previous work has investigated the use of e-paper technology in several application domains. Sánchez-de-Rivera et al. [15] evaluated wireless e-paper displays for retail labeling in supermarket environments, focusing on display performance and user acceptance. Vochin et al. [16] presented a battery-powered wireless e-paper system for public information and emergency alert applications. He et al. [17] studied waveform design for e-paper display, addressing refresh performance, image quality, and ghosting effects. Khan et al. [18] reported an ultra-low-power connected e-paper platform based on Wi-Fi, Message Queuing Telemetry Transport (MQTT), and Long Range (LoRa) communication, capable of operating for more than six months on battery power. While these studies contribute to wireless e-paper systems, display control techniques, and low-power operation, none combine BLE communication, embedded temperature sensing, QR code visualization, and compact PCB-level integration within a single smart label platform.

Table 1 shows that previous studies have primarily focused on wireless e-paper displays, display driving techniques, or infrastructure-oriented information systems. By contrast, the platform presented in this work combines BLE communication, embedded temperature sensing, dynamic QR code rendering, and compact PCB implementation based on the HY0020 SoC. The contribution therefore resides in the integration of these functions within a single low-power smart label architecture rather than in the introduction of a new display or communication technology.

Table 1. Comparative analysis of the main e-paper-based smart labeling and wireless display systems.

Ref.	Application Context	Communication and Sensing	Dynamic E-Paper and QR Support	Compact PCB/Hardware Integration	Processing Platform/SoC	Power/Energy Evaluation	Main Limitation Relative to This Work
[15]	Supermarket/retail information display	Wireless infrastructure; no embedded or temperature sensing	Dynamic e-paper output; QR not central	Partial infrastructure-oriented integration	Not specified	Comparative display evaluation	No BLE SoC-based smart label and no integrated temperature-aware embedded architecture
[16]	Public information/emergency alert display	Wireless display system without sensing	Dynamic e-paper output; no QR-oriented rendering	No clear item-level compact PCB focus	Not specified	Low-power wireless display operation	More signage/emergency-display-oriented than product smart-label-oriented
[17]	Electrophoretic display waveform optimization	No wireless communication or sensing	Display driving and refreshing optimization only	No embedded smart label integration	Not applicable	Indirect display optimization analysis	No communication, sensing, or integrated IoT label architecture
[18]	Ultra-low-power connected e-paper sticky note	Wi-Fi/MQTT and LoRa architecture; no temperature sensing	Dynamic e-paper messaging; not product-traceability-oriented	Partial embedded integration	Generic MCU platform	404 μ A current consumption and >6 months operation	Different communication architecture and not BLE-based nor smart-label-oriented
This work	Item-level BLE smart label for retail, traceability, and DPP applications	BLE 5.4 communication with embedded HY0020 temperature sensing	Dynamic e-paper rendering with QR code support	Compact two-layer PCB designed around a 1.02-inch e-paper module with TXS0108E mixed-voltage interfacing	HY0020 SoC, TXS0108E interface	Joulescope evaluation; 0.772 mA average current and ~54 days estimated autonomy	Monochrome display and autonomy dependent on update/interaction frequency

2.3. Research Gap

The comparison in Table 1 indicates that the main limitation of existing literature is not the absence of e-paper, wireless communication, or low-power sensing as isolated technologies. RFID technology has reached complete development for traceability and identification purposes [5]; BLE provides excellent indoor IoT connectivity solutions that require limited power usage [7]; e-paper technology enables users to see information continuously while consuming minimal power [12]; previous research based on HY0020 already proves the existence of a small energy-saving device that monitors BLE temperature [14]. Existing e-paper studies show that wireless information displays work successfully in retail settings and alert systems and low-power connected messaging applications [15–18]. Although several relevant technologies and systems have been reported in the literature, relatively few studies focus on the practical integration of BLE communication, embedded sensing, dynamic e-paper visualization, and compact PCB-level implementation within a single smart label prototype.

The main gap is therefore not the isolated existence of e-paper, BLE, RFID, or low-power sensing. The actual gap lies in the process of integrating these components into a compact, low-power, reconfigurable label class device which can be used for item identification purposes. The existing literature lacks an integrated solution that combines:

- An ultra-compact BLE SoC platform;
- Embedded temperature sensing;
- E-paper interface capable of persistent dynamic rendering; including QR-oriented content;
- A hardware-level mixed-voltage interfacing stage for reliable display control;
- A custom PCB designed with label-like dimensional constraints and energy-aware subsystem coordination.

The analysis presented in the previous sections indicates that the integration of sensing, wireless communication, and dynamic visual information within compact low-power smart label platforms remains relatively limited in the literature. To explore this design space, a prototype was developed around the HY0020 SoC, integrating temperature sensing, BLE communication, and e-paper visualization on a custom printed circuit board compatible with a 1.02-inch display format. The resulting platform provides a practical example of system-level integration for smart label applications.

3. Materials and Methods

The development of an ultra-low-power smart label requires joint optimization of hardware architecture, embedded firmware, wireless communication, and display management. Unlike conventional IoT nodes, the proposed system must operate under strict dimensional and energy constraints while simultaneously supporting dynamic visual rendering, environmental sensing, and BLE-based interaction. Consequently, the design process cannot be approached as a simple combination of independent commercial modules; instead, it requires a hardware–firmware co-design methodology in which sensing, communication, and display subsystems are coordinated according to power-aware operational strategies.

This section describes the materials, implementation methodology, firmware architecture, and experimental procedures adopted in the development of the proposed smart label platform. First, the hardware architecture and PCB integration are presented, including the HY0020 SoC, TXS0108E mixed-voltage interface, external memory [19], and e-paper display subsystem. Next, the firmware organization implemented in Zephyr’s [20] Real-Time Operating System (RTOS) environment is detailed, with emphasis on task separation, BLE interaction management, and update scheduling for the electrophoretic display. Finally, experimental scenarios used to validate system functionality, sensing reliability, and energy consumption are described.

3.1. System Implementation and Hardware Setup

Figure 2 illustrates the hardware architecture of the developed smart label, designed for low-power IoT edge applications. This platform is centered on the HY0020 SoC, which provides BLE connectivity together with processing and control functions. Temperature monitoring is performed using the integrated sensing capability of the device, enabling environmental supervision in applications such as logistics and cold-chain monitoring.

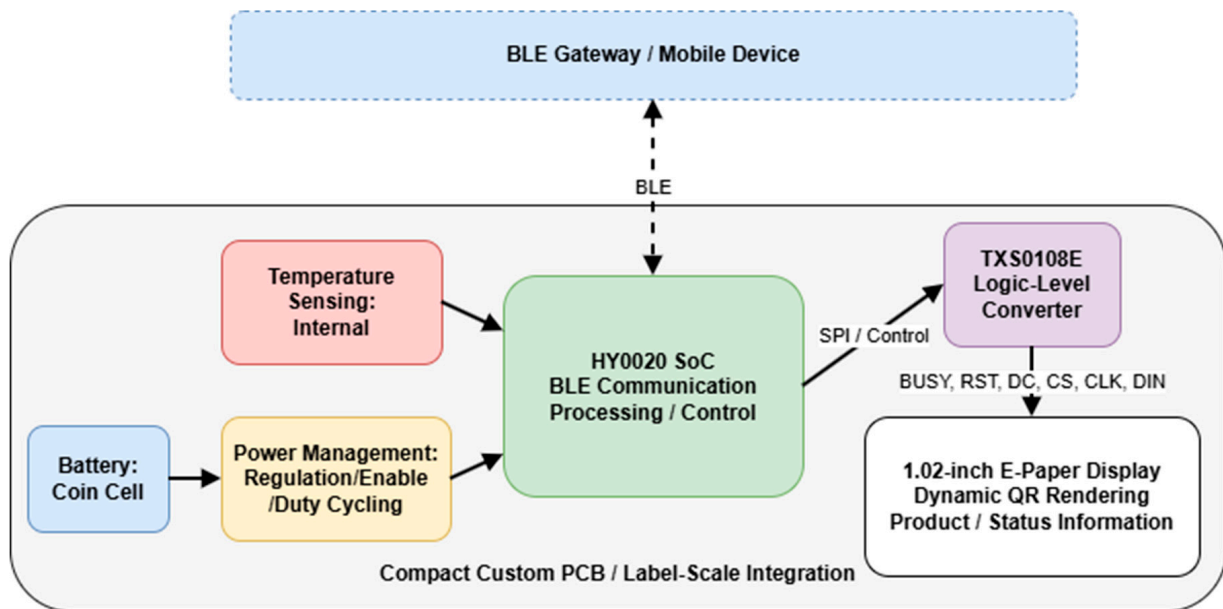


Figure 2. Block diagram of the proposed low-power smart label architecture, integrating the HY0020 SoC, BLE communication, temperature sensing, TXS0108E level shifting, power management, and a 1.02-inch e-paper display for dynamic QR and status rendering.

Power is supplied by a coin-cell battery and managed through a dedicated power management stage responsible for voltage regulation and duty cycling strategies aimed at reducing energy consumption. Dynamic information is presented on a 1.02-inch e-paper module capable of displaying QR codes and status information while maintaining extremely low standby power requirements.

Communication between the HY0020 and the display is performed through an SPI-based interface comprising BUSY, RST, DC, CS, CLK, and DIN control signals. A TXS0108E device was incorporated to improve interface robustness and compatibility. Although both devices operate from a nominal 3 V supply, the level translator provides additional tolerance to signal-level variations, supports future compatibility with alternative display modules, and contributes to reliable operation of the display control interface.

The hardware layout was developed around the physical dimensions of the e-paper module, resulting in a compact printed circuit board that combines communication, sensing, power management, and visual information display within a single embedded platform.

Figure 3 presents a simplified functional representation of the proposed hardware architecture. The HY0020 SoC acts as the central processing unit, integrating BLE communication capabilities and an internal temperature sensor. Display communication is performed through the TXS0108E (no comparative reliability measurements with and without the TXS0108E were performed in the current study) interface, which provides robust signal conditioning and compatibility with respect to the SPI-based e-paper module. Power is supplied by a CR2477 lithium coin-cell battery and managed through the dedicated power-management circuitry.

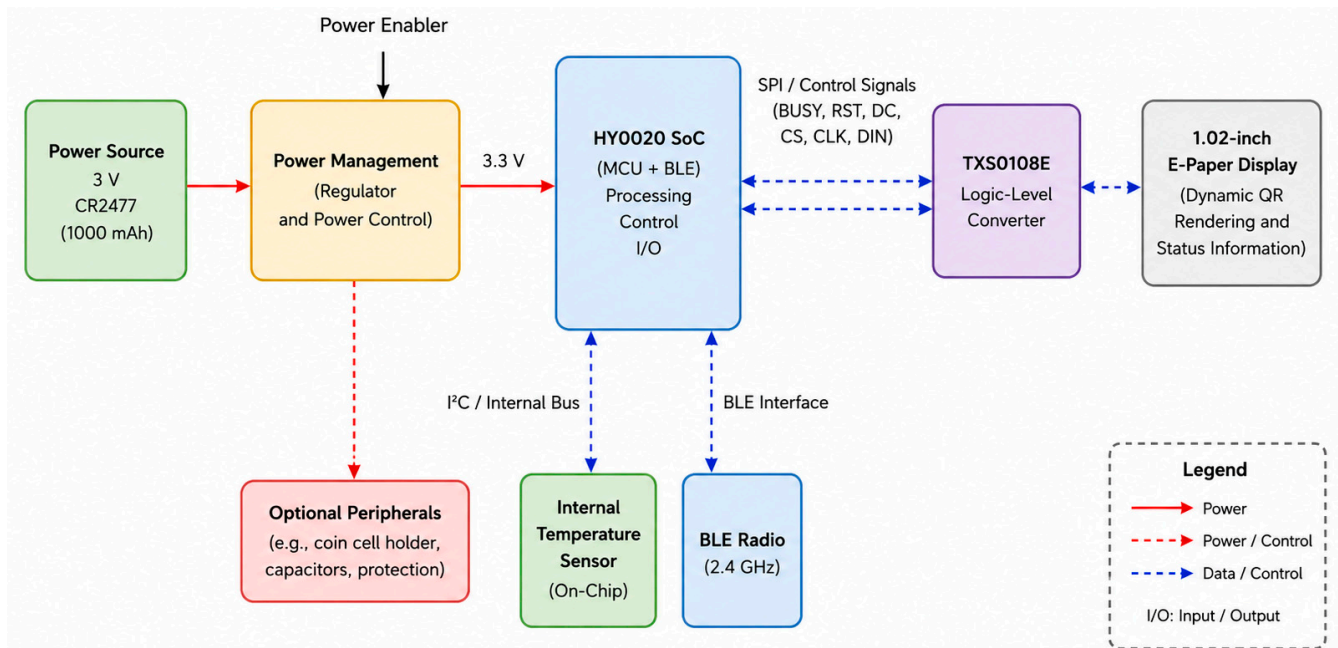


Figure 3. Simplified hardware architecture of the proposed smart label platform, highlighting the interaction between the HY0020 SoC, integrated BLE communication, internal temperature sensor, TXS0108E logic-level interface, e-paper display module, and power management subsystem.

As shown in Figure 3, the architecture was designed to minimize hardware complexity while maintaining support for wireless communication, temperature monitoring, and dynamic information visualization. This functional partition simplifies integration and supports low-power operation suitable for smart label applications.

3.2. Hardware Integration

Communication between the HY0020 SoC and the e-paper module is performed through a dedicated control and data interface routed via the TXS0108E level translator. The interface includes the SPI clock and data signals, together with display control and status monitoring lines required for display initialization, refresh management, and command execution. Table 2 summarizes the functional mapping between the HY0020 interface, the TXS0108E translator, and the corresponding e-paper control signals.

Table 2. Functional mapping between the HY0020 interface, TXS0108E level translator, and e-paper display control signals.

HY0020 Interface	TXS0108E Channel	E-Paper Signal	Function
SPI Clock	A1/B1	CLK	Display clock
SPI Data	A2/B2	DIN	Serial display data
GPIO Output	A3/B3	CS	Chip selection
GPIO Output	A4/B4	DC	Data/Command selection
GPIO Output	A5/B5	RST	Display reset
GPIO Input	A6/B6	BUSY	Display status monitoring

The signal mapping shown in Table 2 corresponds to the integration adopted in the prototype and is consistent with the electrical schematics presented in Figures 4 and 5. The TXS0108E interface provides a dedicated adaptation stage for the CLK, DIN, CS, DC, RST, and BUSY signals exchanged between the HY0020 SoC and the e-paper display subsystem.

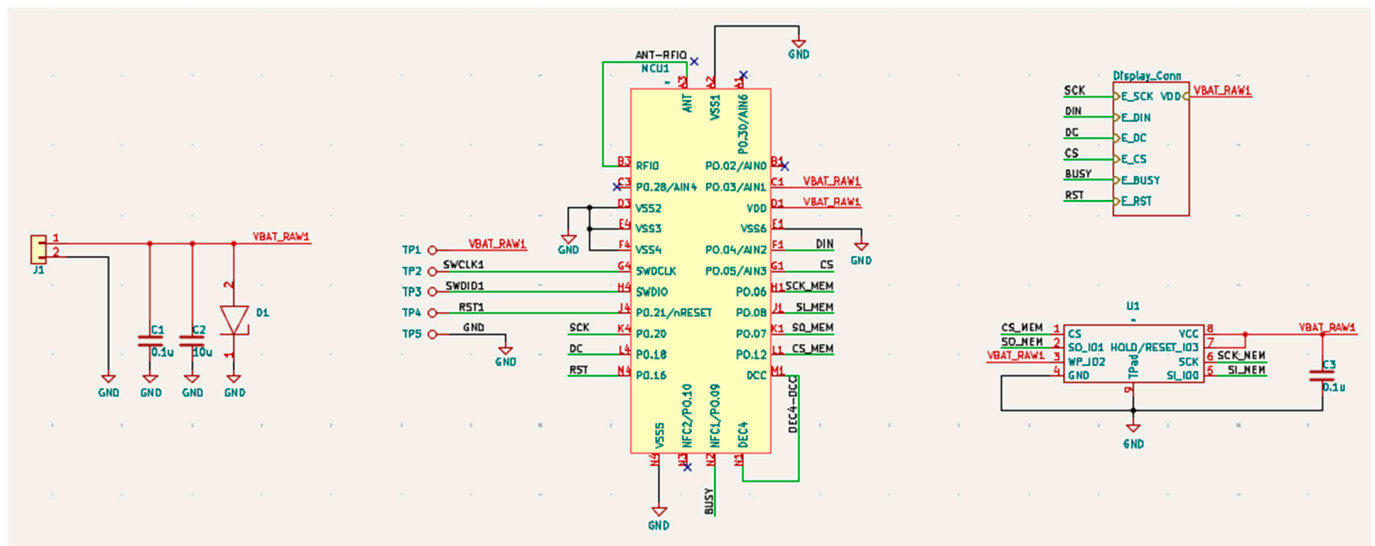


Figure 4. Electrical schematic of the proposed smart label platform HY0020 SoC subsystem, external memory, and power circuitry.

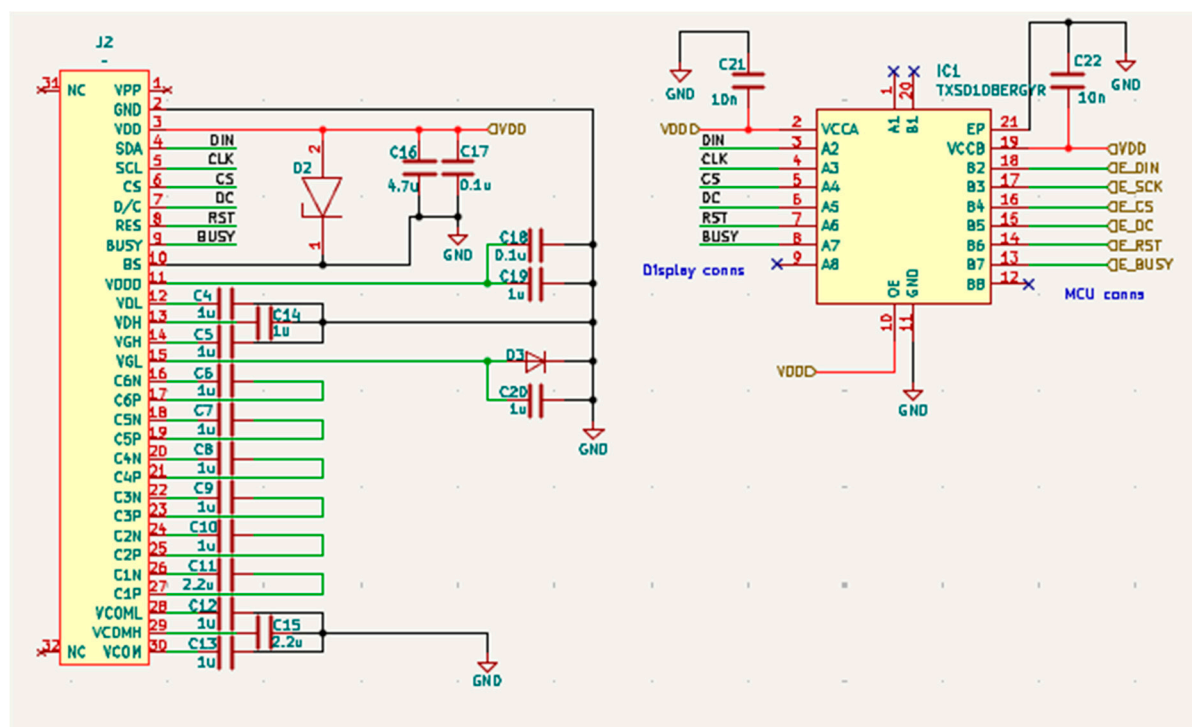


Figure 5. Electrical schematic of the proposed smart label platform e-paper display interface and TXS0108E signal conditioning stage.

As shown in Figure 6, the TXS0108E was positioned close to the display connector to reduce control signal trace lengths and simplify routing between the HY0020 and the e-paper module. During experimental operation, no communication errors or display control failures were observed. All the previously mentioned hardware components (HY0020 SoC, TXS0108E logic-level converter, and coin cell holder) were integrated into the PCB layout, shown in Figure 4. Additionally, a Flexible Printed Circuit (FPC) connector for the e-paper display and an ATX25XE161D 16 Mbit external flash memory [19] were added to the design. The PCB layout (with two physical layers) was constrained by physical dimensions of a 1.02-inch e-paper module, making the display footprint the dominant

mechanical design reference. This approach supports label-scale integration rather than a generic development-board implementation. The PCB has $3.8 \times 1.8 \times 0.7$ cm dimensions without the e-paper display (Figures 7 and 8), and $3.8 \times 1.9 \times 1$ cm when the display is installed (Figure 9).

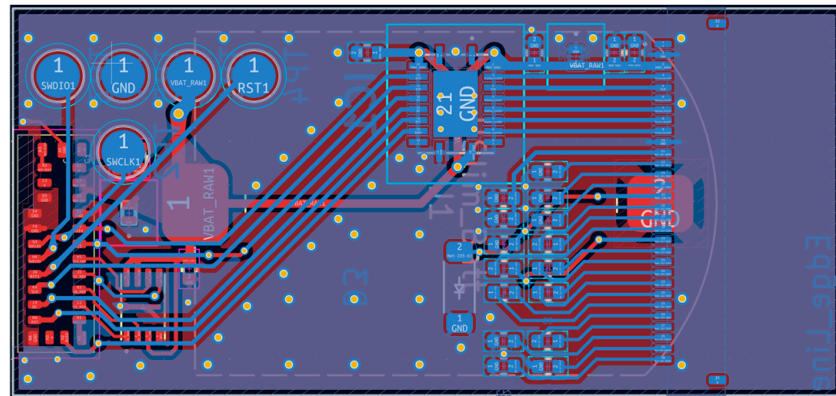


Figure 6. PCB layout of the smart label.



Figure 7. Smart label PCB dimensions—Width.



Figure 8. Smart label PCB dimensions—Length and Height.

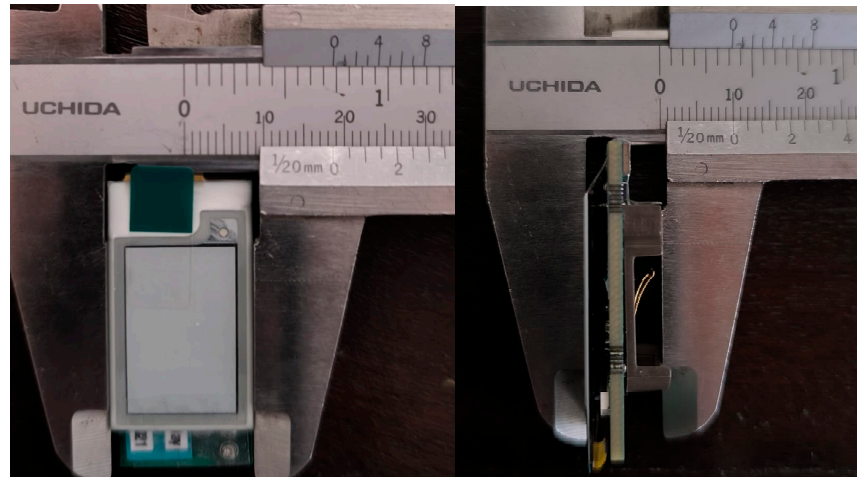


Figure 9. Smart label with e-paper display—Width and Height.

Additionally, as shown in Figure 10, an enclosure with outer dimensions of $7 \times 2.7 \times 1.2$ cm was also designed to accommodate the smart label.

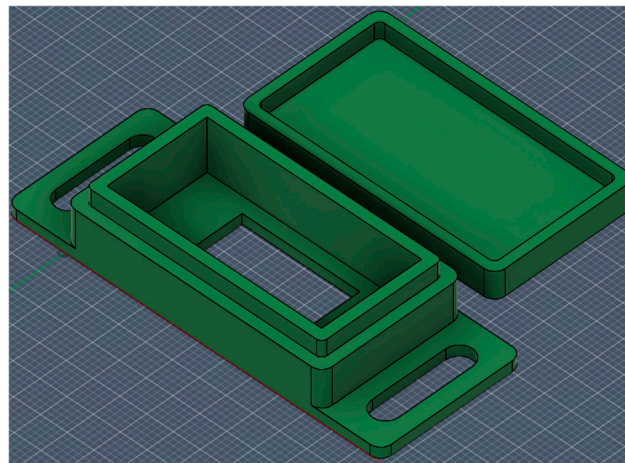


Figure 10. Smart label enclosure 3D model.

The firmware is organized around two main operating modes: Standalone Mode and BLE/User Mode. Standalone Mode is responsible for periodic display updates and temperature presentation, whereas BLE/User Mode manages wireless interactions, user authentication, QR code visualization, and timeout handling. The overall architecture was designed to minimize processing overhead while maintaining responsiveness during user interactions.

3.3. Algorithms Developed

An example firmware was developed to illustrate the functionalities of the smart label, as depicted in the flowchart presented in Figure 11. Firmware was implemented using the C programming language within the Zephyr [20] RTOS environment. This example assumes the use of the smart label in a food retail facility, where its primary function is to display information about a specific product (e.g., product name, price, and product code) and the ambient temperature surrounding the product. Additionally, the label can communicate via BLE with a mobile device equipped with, for example, a retail facility app. This allows it to display personalized user information and scannable QR codes on the label itself, which links to product details such as traceability information or promotions.

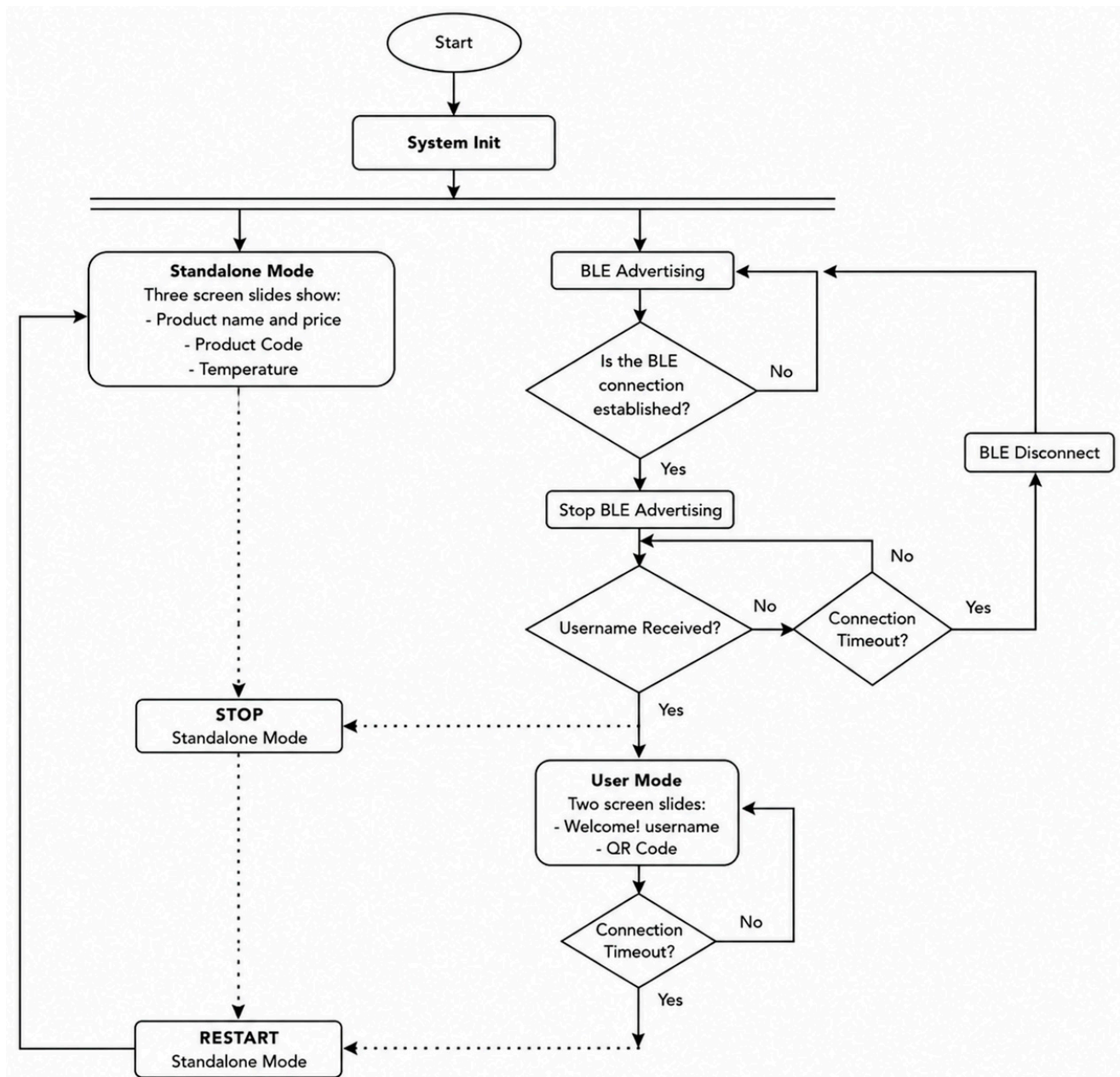


Figure 11. Smart label firmware flowchart.

The procedure starts as described in Algorithm 1 with the activation and configuration of all the necessary elements such as the HY0020 SoC, e-paper display, and BLE interface. Following system initialization, two distinct threads are created. The first thread is responsible for cyclically updating the e-paper display with all relevant information regarding a given product, as well as the surrounding ambient temperature, measured by SoC internal sensor. This operational state is referred to as Standalone Mode. The second thread manages the BLE interface and is responsible for displaying personalized user information, as well as traceability and promotional QR codes on the e-paper display. This operational state is referred to as User Mode. All product-related screens and QR codes are preloaded into non-volatile memory and accessed by the HY0020 SoC during runtime. In this example, Standalone Mode, presented in Algorithm 2, is responsible for updating three distinct screens on the e-paper display. The first presents the product name and price, the second displays the product code, and the third shows the surrounding ambient temperature. The display duration of each screen is configurable and, in this case, was

set to 10 s. Temperature measurement is performed immediately prior to updating the corresponding temperature screen. By leveraging e-paper technology and its ability to retain the last displayed content without continuous power supply, the display is turned off after each update. This significantly reduces the overall system power consumption when compared to more conventional LCD or OLED displays. In the absence of any BLE interaction, the system continuously cycles through this mode.

Algorithm 1. Smart Label Firmware Workflow

```

1: Initialize HY0020 SoC
2: Initialize BLE communication
3: Initialize e-paper display
4: Start Standalone Mode
5: Start BLE advertising (250 ms interval)
6: while device is powered do
7:   if BLE connection established then
8:     Stop BLE advertising
9:     Wait for username
10:    if username received then
11:      Display welcome message
12:      Load QR bitmap
13:      Display QR bitmap
14:      Enter User Mode
15:    end if
16:    if connection timeout (1 min) then
17:      Disconnect BLE
18:      Restart BLE advertising
19:    end if
20:  else
21:    Execute Standalone Mode:
22:    - Product Name and Price
23:    - Product Code
24:    - Temperature
25:  end if
26: end while

```

Algorithm 2. Standalone Mode Thread

```

1: Load display content
2: while active do
3:   Display product name and price
4:   Wait 10 s
5:   Display product code
6:   Wait 10 s
7:   Read temperature sensor
8:   Display temperature
9:   Wait 10 s
10: end while

```

The BLE/User Mode thread is initialized by enabling BLE advertising with a 250 ms periodic interval, remaining active until a connection request is received from a mobile device. Once connected, advertising is disabled to avoid simultaneous connections. The thread then enters a waiting state, expecting a username string (e.g., linked to a retail facility app account). If no data is received within 1 min, the system automatically closes the connection. Following data reception, the execution of the Standalone Mode thread is suspended to allow the BLE/User Mode thread to access the e-paper display. Once access is established, the message “Hello, username!” is displayed, followed by presentation of a scannable QR code after a 10 s interval. The thread then remains in a waiting state until either a 1 min timeout is reached or a disconnection occurs from the mobile device, after which the Standalone Mode thread and BLE advertising are resumed. Algorithms for the BLE/User Mode thread can be observed in Algorithm 3.

Algorithm 3. BLE/User Mode Thread

```

1: Start BLE advertising
2: while active do
3:   Wait for BLE connection
4:   if connection established then
5:     Receive username
6:     Suspend Standalone Mode
7:     Display username
8:     Retrieve and display QR bitmap
9:     Wait for timeout or user disconnection
10:    Resume Standalone Mode
11:   end if
12: end while

```

Although the system functions and logic are highly configurable and adaptable to various scenarios, the algorithms previously described effectively illustrate different functionalities offered by the system. In the implemented prototype, BLE advertising was configured with a 250 ms interval. User interaction was based on a single active BLE connection, with a 1 min timeout period. The expected payload format was “1Username\n”, followed by transmission of an acknowledgment message. The current implementation supports a single-user interaction model and does not yet address large-scale multi-label coordination.

Display updates are managed through an event-driven scheduling strategy. During standalone operation, the system cycles through the predefined information screens according to the configured timing parameters. When a BLE connection is established, the standalone display cycle is suspended, and screen updates are performed only when new information is received or when the displayed content requires modification. This approach reduces unnecessary display refresh operations, limits energy consumption, and decreases the likelihood of ghosting accumulation caused by repeated electrophoretic transitions.

The interaction between the mobile user device and the smart label platform is illustrated in Figure 12. The communication process begins with BLE advertising by the smart label, followed by device discovery, connection establishment, username transmission, acknowledgment exchange, dynamic display generation, and automatic session termination after a predefined timeout period.

The implemented communication workflow supports a single active BLE session at a time. After receiving the user identifier, the smart label updates the e-paper display with the corresponding content and remains available until the timeout condition is reached, after which the system automatically disconnects and returns to advertising mode.

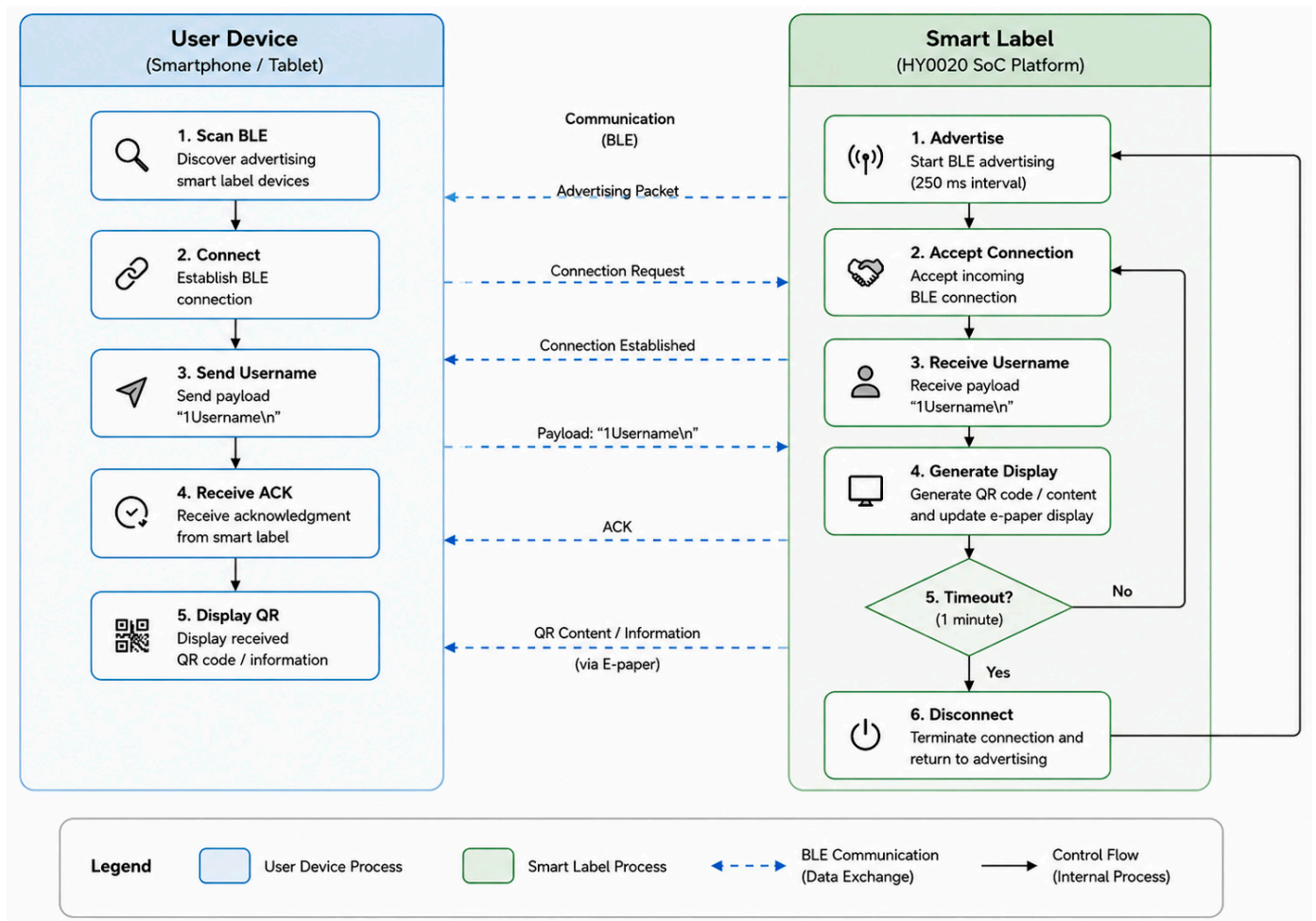


Figure 12. Firmware interaction workflow between the user device and the smart label platform, including BLE discovery, connection establishment, username transmission, acknowledgment handling, display update, timeout management, and disconnection procedures.

Power management follows an event-driven strategy. The display remains inactive between refresh operations, BLE activity is restricted to advertising and short interaction sessions, and temperature measurements are performed only when required for display updates. Consequently, processing activity remains low during normal operation. Firmware memory usage is primarily associated with BLE communication routines, display control functions, and bitmap storage, while the use of pre-generated QR images avoids additional computational load related to real-time QR code generation.

4. Results

This section presents the experimental validation of the proposed smart label platform. The evaluation is organized into six parts: BLE communication and functional validation, temperature sensing assessment, power consumption characterization, battery lifetime estimation under different operating profiles, QR code readability testing, and ghosting evaluation after repeated display refresh operations.

4.1. BLE Communication and Functional Validation

The first experiment evaluated the functional operation of the smart label prototype and the BLE-based interaction workflow described in Section 3.3. Figure 13 shows the three screens used in standalone mode, corresponding to product information, product code,

and ambient temperature. These screens were designed to provide direct visual feedback while maintaining a simple layout suitable for the 1.02-inch e-paper display.

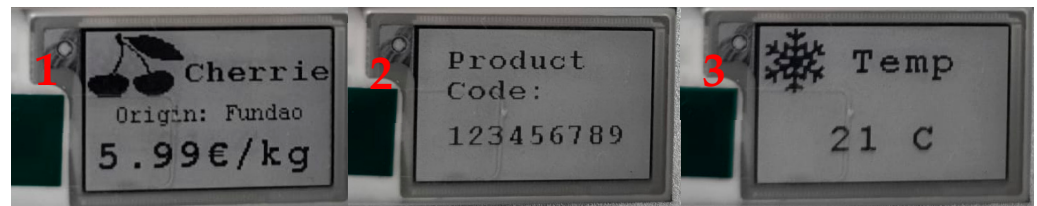


Figure 13. Screens included in Standalone Mode. These consist of the product name, origin, and price (1), a generic product code (2), and the ambient temperature (3) measured by the SoC internal sensor.

BLE/User Mode was tested using the Serial Bluetooth Terminal application [21,22]. As shown in Figures 14 and 15, the mobile device detected the BLE advertising packet, established a connection with the smart label, transmitted the user string, and received an acknowledgment message from the device.

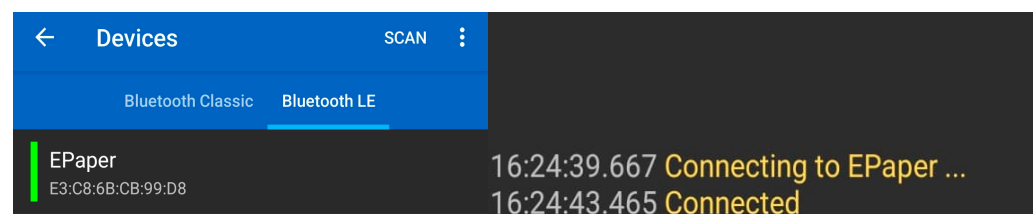


Figure 14. Scanning and connection request by Serial Bluetooth Terminal app.

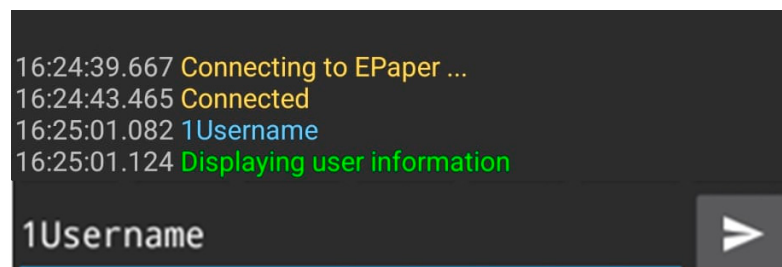


Figure 15. Data sent by Serial Bluetooth Terminal app and ACK message received from the smart label.

Once the connection is established, a 1 min timeout is initiated, during which the mobile device sends required data to the smart label. In this example, the smart label expects to receive a string in the format “1Username\n”. Upon reception, the information contained between identifier “1” and the newline character is extracted and stored, and an acknowledgment (ACK) message is sent back to the mobile device, as shown in Figure 15.

After reception of the user data, the smart label suspended the standalone display cycle and updated the e-paper display with the corresponding welcome message and QR code screen, as illustrated in Figure 16. The QR code was generated offline during content preparation and stored as a bitmap in external memory. During operation, the HY0020 retrieves the selected bitmap and transfers it to the e-paper controller, avoiding the computational and memory overhead associated with real-time QR code generation.

The complete interaction sequence is shown in Figure 17. After the BLE exchange is completed, the connection is terminated either by the mobile device or by the timeout mechanism implemented in the firmware. The system then resumes BLE advertising and returns to standalone operation.

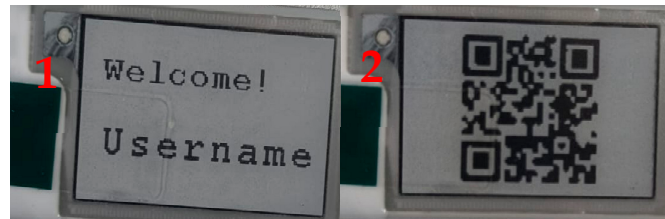


Figure 16. Screens included in BLE/User Mode. This consists of a welcome message (1), followed by an example of a scannable QR code (2).

```

16:24:39.667 Connecting to EPaper ...
16:24:43.465 Connected
16:25:01.082 1Username
16:25:01.124 Displaying user information
16:26:01.249 Connection lost
    
```

Figure 17. Complete interaction from the perspective of the mobile device/app. This interaction sequence includes connection establishment with the smart label, transmission of user data, reception of ACK message, and automatic termination of the BLE connection by the smart label upon timeout.

4.2. Temperature Sensing Evaluation

The HY0020 integrated temperature sensor was evaluated through a 48 h outdoor ambient test using a DHT22 [22] sensor as a low-cost reference. Since the DHT22 has its own finite accuracy tolerance, the experiment should not be interpreted as a metrological calibration of the HY0020 sensor. Instead, it provides a practical assessment of relative temperature tracking and measurement stability under representative ambient conditions.

Figure 18 shows the temperature evolution measured by both sensors during the 48 h period. The two curves follow the expected daily thermal variation, with temperatures increasing during daytime and decreasing during nighttime.

Figure 19 presents the corresponding error between the HY0020 sensor and the DHT22 reference.

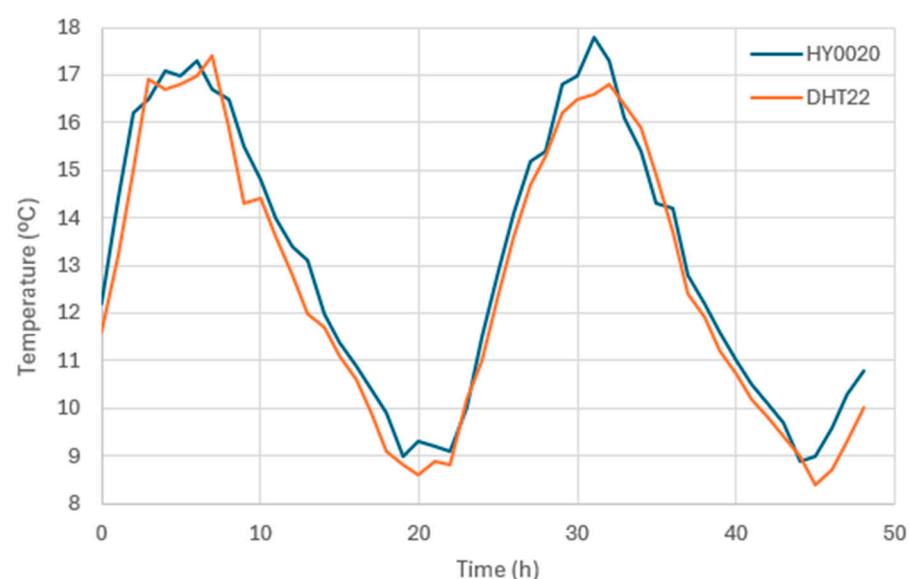


Figure 18. Outdoor ambient temperature measured by the HY0020 SoC temperature sensor and DHT22 during a 48 h period.

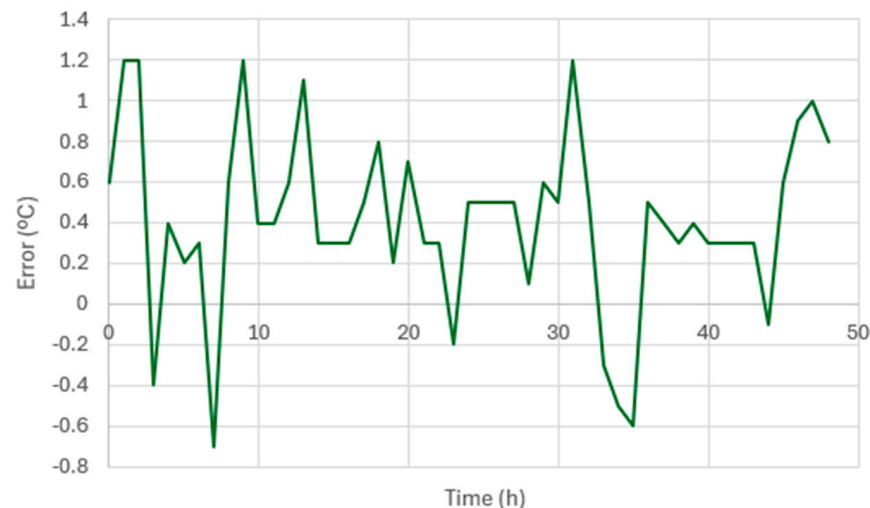


Figure 19. Temperature error between the HY0020 internal sensor and DHT22 reference sensor.

The HY0020 sensor exhibited a mean error of 0.410°C , a mean absolute error of 0.524°C , and a standard deviation of 0.442°C relative to the DHT22 reference. These values indicate stable relative tracking under the evaluated environmental conditions. However, since DHT22 was used only as a low-cost reference, the reported metrics should not be interpreted as absolute accuracy values. A controlled calibration using a laboratory-grade reference sensor and temperature chamber would be required to determine the absolute metrological performance of the HY0020 sensor.

4.3. Power Consumption Characterization

Energy consumption measurements of the smart label were conducted using a Joulescope JS220 energy analyzer, Jetperch LLC, Rockville, MD, USA [23] over a full execution period, with obtained results presented in Figure 20. A complete execution cycle was defined as the display of three screens associated with Standalone Mode and BLE advertising, followed by establishment of a BLE connection between the smart label and a mobile device, corresponding to user interaction and subsequent termination of BLE connection upon timeout, after which the system resumes Standalone Mode and BLE advertising.

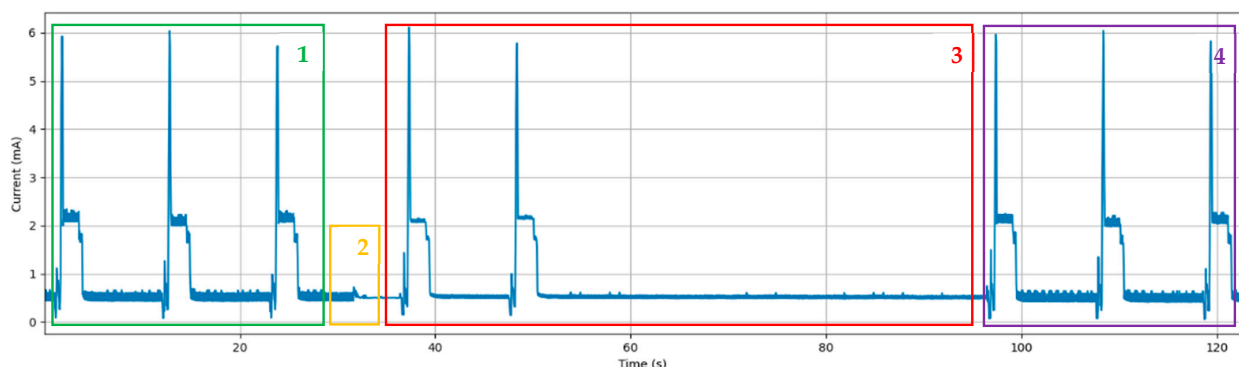


Figure 20. Energy consumption recorded during a full execution cycle. This cycle comprises the presentation of three standalone mode screens and BLE advertising (1), BLE connection establishment with a mobile device (2), presentation of user-related screens (3), and termination of BLE connection followed by resumption of Standalone Mode and BLE advertising (4).

Figure 20 allows the identification of different firmware execution phases mentioned above. Among them, most prominent feature is the occurrence of current peaks around 6 mA whenever the e-paper display updates, corresponding to the highest power con-

sumption period of the smart label. Another relevant feature to highlight is the reduction in standby power consumption after establishment of a BLE connection. This behavior is expected due to the interruption of continuous BLE advertising once a connection is established.

The current profile shown in Figure 20 indicates that e-paper refresh operations represent the dominant instantaneous power events, reaching peak currents close to 6 mA. By contrast, BLE communication and temperature sensing operations contribute less significantly to the overall energy budget due to their short duration and relatively low duty cycle. The measured operating cycle resulted in an average current consumption of 0.772 mA, which was subsequently used for the battery lifetime estimations presented in Table 3. Table 3 summarizes the approximate current levels associated with each operating phase identified in Figure 20.

Table 3. Approximate current consumption characteristics of the main operating phases estimated from the profile shown in Figure 20.

Operating Phase	Approximate Current Level	Typical Duration	Observation
BLE advertising	0.5–0.7 mA	Continuous between events	Low-power idle communication
BLE connection and data exchange	0.5–0.8 mA	Few seconds	Short user interaction period
Temperature sensing	<1 mA	Very short	Negligible contribution to total energy budget
E-paper refresh	Peak $\approx$ 6 mA	$\sim$ 2 s	Dominant instantaneous power event
Full operating cycle	Average 0.772 mA	120 s measured cycle	Basis for autonomy estimation

A critical requirement for assessing the suitability of the proposed smart label for real-world deployment is the estimation of its operational autonomy. Assuming continuous execution of the previously described cycle, which presents an average current draw of 0.772 mA, and considering the use of a 3 V CR2477 lithium battery with a nominal capacity of 1000 mAh, the smart label is expected to operate continuously for approximately 54 days. It is important to note that actual power consumption may vary depending on the number of BLE interactions performed by the smart label.

A quantitative summary of the principal experimental results is presented in Table 4. Reported metrics include a current consumption profile of the smart label, estimated battery-powered operational lifetime, and statistical indicators associated with the accuracy and stability of the HY0020 internal temperature sensor.

Table 4. Summary of principal experimental results obtained for the proposed BLE smart label, including power consumption characteristics, estimated operational autonomy, and statistical temperature sensing accuracy metrics.

Parameter	Measured/ Estimated Value	Relevance
E-paper update peak current	$\sim$ 6 mA	Identifies display refresh as dominant instantaneous load
Average current during full cycle	0.772 mA	Basis for autonomy estimation

Table 4. *Cont.*

Parameter	Measured/ Estimated Value	Relevance
Battery capacity	1000 mAh	CR2477 reference capacity
Estimated autonomy	~54 days	Practical battery-powered operation
Mean temperature error	0.410 °C	Sensor bias
Mean absolute error	0.524 °C	Average deviation from DHT22
Error standard deviation	0.442 °C	Measurement stability

4.4. Battery Lifetime Assessment Under Different Operating Profiles

To evaluate the influence of operating conditions on autonomy, three representative profiles were measured by varying BLE interaction frequency and display refresh activity. The results, summarized in Table 5, show that display refresh activity and BLE interaction frequency are the dominant factors affecting battery lifetime.

Table 5. Estimated battery lifetime under representative operating profiles.

Scenario	Operating Conditions	Average Current (mA)	Estimated Lifetime (Days)
Low Activity	Reduced BLE interactions, periodic display updates	0.830	50.2
Medium Activity	Experimental operating profile	0.772	54.0
High Activity	Frequent BLE interactions and QR updates	1.416	29.4

The medium-activity scenario resulted in a slightly lower average current than the low-activity scenario because the BLE timeout period suspends display cycling, reducing the number of refresh operations during the measured interval. The operating assumptions used for the autonomy estimate are summarized in Table 6.

Table 6. Operating profile used for autonomy estimation.

Parameter	Value
BLE Advertising Interval	250 ms
BLE Timeout	60 s
Number of Standalone Screens	3
Screen Duration	10 s
E-paper Refresh Time	~2 s

Table 6. *Cont.*

Parameter	Value
Temperature Sampling	Once per temperature screen
Active BLE Connections	1
Battery Type	CR2477 (1000 mAh)

4.5. QR Code Readability Assessment

Figure 21 illustrates representative QR code visualization under indoor illumination and direct sunlight.



Figure 21. QR code visualization under typical indoor illumination (**left**) and direct sunlight conditions (**right**).

A preliminary QR code readability assessment was performed under different viewing distances, observation angles, illumination conditions, and display refresh states. The results are summarized in Table 7. Successful decoding was achieved in all frontal-view tests between 10 cm and 30 cm and under direct sunlight. However, readability decreased significantly with increasing viewing angle and under low light conditions. No degradation in QR code readability was observed after repeated display refresh operations. Each condition was evaluated using ten independent scanning attempts.

Table 7. QR code readability assessment under different operating conditions.

Test Condition	Successful Scans/Attempts
10 cm, 0°	10/10
20 cm, 0°	10/10
30 cm, 0°	10/10
20 cm, 30°	3/10
20 cm, 45°	0/10
Low light environment	6/10
Direct sunlight	10/10
After repeated refreshes	10/10

4.6. Ghosting Evaluation After Repeated Refresh Operations

The effect of repeated full refresh operations was evaluated to assess visible ghosting and QR code readability. A preliminary ghosting assessment was conducted after 1, 10, 25,

and 50 consecutive full refresh operations. As shown in Table 8, only low levels of visible ghosting were observed, with no impact on QR code readability.

Table 8. QR readability and ghosting assessment after repeated display refresh operations.

Number of Refreshes	QR Readable	Visible Ghosting Level
1	Yes	None
10	Yes	Very Low
25	Yes	Low
50	Yes	Low

To complement the qualitative ghosting assessment presented in Table 8, Figure 22 shows an example of the display after 50 consecutive full refresh operations. The image illustrates the presence of minor residual artifacts in the background region while preserving adequate QR code contrast and readability.



Figure 22. QR code displayed after 50 consecutive full refresh operations. Minor residual ghosting artifacts can be observed in the background region; however, the QR code remained fully readable during all decoding tests.

Visual inspection revealed low levels of residual ghosting after repeated refresh operations. The observed artifacts were mainly confined to the display background and did not significantly affect QR code contrast or decoding performance. All QR code scans remained successful after 50 consecutive refresh cycles, indicating that the adopted refresh strategy provides adequate robustness for the evaluated operating conditions.

5. Discussion

The experimental results confirm that the proposed platform integrates BLE communication, temperature sensing, and e-paper visualization within a compact low-power smart label architecture. The BLE tests demonstrated bidirectional interaction with a mobile device, while the QR code workflow confirmed that pre-generated bitmap rendering can reduce processing requirements on the HY0020 SoC.

The temperature experiment showed stable relative tracking between the HY0020 internal sensor and the DHT22 reference, although the results should not be interpreted as metrological calibration. Power measurements confirmed that e-paper refresh events dominate instantaneous current consumption, while the estimated autonomy depends strongly on display refresh frequency and BLE interaction rate.

The QR code readability tests showed reliable frontal scanning between 10 cm and 30 cm and under direct sunlight, but performance decreased at larger viewing angles and under low light conditions. The repeated refresh test also showed low residual ghosting after 50 full refreshes, without loss of QR readability.

Recent DPP frameworks and connected product traceability systems reinforce the relevance of low-power smart labels capable of combining sensing, wireless communication, and dynamic visual information [24,25]. In parallel, intelligent-sensing research is moving toward compact platforms that combine sensing, communication, local processing, and energy-efficient operation [26–28]. Although the proposed system does not implement advanced sensing intelligence, it provides a practical foundation for future extensions involving partial refresh, multi-label BLE coordination, color e-paper, energy harvesting, and additional sensing modalities.

6. Conclusions

This work presented the design and experimental validation of a compact smart label platform based on the HY0020 SoC, integrating BLE communication, temperature sensing, and e-paper visualization within a single embedded architecture. The developed prototype demonstrated reliable BLE connectivity, dynamic QR code rendering, and low-power operation through a custom PCB specifically designed for smart label applications.

The experimental results confirmed the feasibility of combining sensing, wireless communication, and e-paper display technology in a battery-powered platform. The bistable nature of the e-paper display enabled persistent visual information with minimal energy consumption between refresh operations, while the integrated temperature sensor exhibited stable behavior throughout the evaluation period. The measured operating profile also provided insight into the interaction between display activity, wireless communication, and overall energy consumption. Additional QR code tests showed reliable frontal readability under representative lighting conditions, while repeated full refresh operation produced only low residual ghosting without preventing QR code decoding.

The proposed platform represents a practical example of system-level integration involving sensing, communication, display management, and power-aware operation within a compact embedded device. Although the current implementation was developed as a proof-of-concept prototype, the results demonstrate the viability of the proposed architecture for smart label applications requiring dynamic visual information and low-power wireless interaction.

Future developments will focus on reducing display update latency through partial refresh techniques, extending BLE networking capabilities, and evaluating additional sensing and energy harvesting functionalities.

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Abbreviations

The following abbreviations are used in this manuscript:

ACK	Acknowledgment
BLE	Bluetooth Low Energy
DPP	Digital Product Passport
FPC	Flexible Printed Circuit
IoT	Internet of Things
LCD	Liquid Crystal Display
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
MQTT	Message Queuing Telemetry Transport
NB-IoT	Narrowband Internet of Things
OLED	Organic Light-Emitting Diode
PCB	Printed Circuit Board
QR	Quick Response
RFID	Radio-Frequency Identification
RSSI	Received Signal Strength Indicator
RTOS	Real-Time Operating System
SDGs	Sustainable Development Goals
SoC	System-on-Chip
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver/Transmitter

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